

G.L. BAJAJ INSTITUTE OF TECHNOLOGY & MANAGEMENT
GREATER NOIDA
B.Tech (CSDS, CSIT, CSE, CS-H, CS-AIML) 2nd Year
Sessional Test-1 (ODD SEM 2026-27)

ROLL NO.

Subject Name: Digital Electronics

Duration: 2:00 Hrs

Subject Code: BOE310

Max. Marks: 50

- Note:
- (i) No student will be allowed to leave the examination Room before end of the exam.
 - (ii) Diagrams should be neat and clean.
 - (iii) Mention question number/section correctly.
 - (iv) Be precise in your answer.
 - (v) Do not write anything on question paper except Roll number.

Course Outcomes:

Following are the course outcomes of the subject :-

CO Code	Course Outcome (CO)	Bloom's Level
BOE310.1	Apply Concepts of Digital Binary Systems and Implementations of Gates.	Applying(K3)
BOE310.2	Analyze and design of Combinational Logic Circuits.	Applying(K3), Analyzing(K4)
BOE310.3	Analyze and design of Sequential Logic Circuits with their applications.	Applying(K3), Analyzing(K4)
BOE310.4	Implement the design procedure of Synchronous & Asynchronous Sequential Circuits.	Applying(K3)
BOE310.5	Apply the concept of Digital Logic Families with Circuit implementation.	Applying(K3)

Section: A

1. Attempt all questions:

Q. No.	Questions	Marks	CO	BL
			(5*2 = 10)	
a)	Represent the -9 as a signed binary number in 2's complement system.	2	BOE310.1	K3
b)	Explain the difference between SOP and POS form.	2	BOE310.1	K2
c)	Define the term universal gates and their applications	2	BOE310.1	K2
d)	Represent the unsigned decimal numbers 965 and 672 in BCD and show the steps necessary to find their sum.	2	BOE310.2	K2
e)	What is the main application of latches and flip-flops in digital systems?	2	BOE310.3	K2

Section: B

2. Attempt any four of the following:

Q. No.	Questions	Marks	CO	BL
			(4*5 = 20)	
a)	Design an XOR gate using NAND gate implementations.	5	BOE310.1	K3
b)	Realize the expression F using an 4-to-1 multiplexer $F = f(x,y,z) = \Sigma(1,2,4,5,7)$	5	BOE310.1	K3
c)	Design an explain 4 bit parallel adder/subtractor circuit.	5	BOE310.2	K3
d)	Explain the working of a BCD adder. How does it differ from a binary adder?	5	BOE310.2	K2
e)	Explain and Design the NAND based S-R Flip flop with truth table.	5	BOE310.3	K3
f)	Explain the operation of a T flip-flop with its truth table and characteristic table.	5	BOE310.3	K2

Section: C

3. Attempt any one question:

Q. No.	Questions	Marks	CO	BL
			(1*10 = 10)	

a)	Solve and minimize using K-map $Y = \sum m(0, 2, 4, 9, 11, 13, 17, 21, 25, 29, 31) + \sum d(6, 15, 27)$ and implement it with AND-OR and NAND-NAND realization.	10	BOE310.1	K3
b)	Simplify the logic function given below using the Quine McCluskey minimization technique, and realize the simplified expression using universal gates. $F(A, B, C, D) = \sum m(0, 1, 3, 7, 8, 9, 11, 15)$	10	BOE310.1	K3

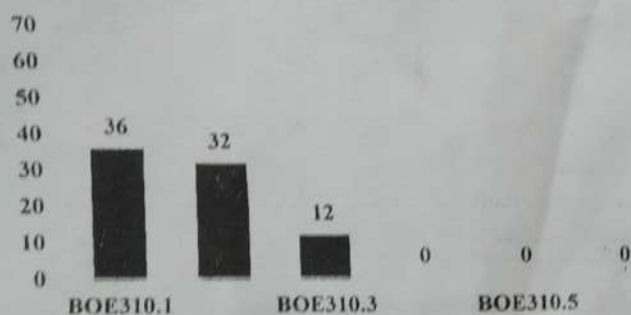
Section: D

4. Attempt any one question:

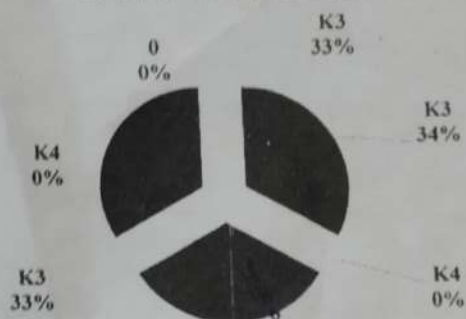
(1*10 = 10)

Q. No.	Questions	Marks	CO	BL
a)	Explain and Design 3-bit magnitude Comparator.	10	BOE310.2	K3
b)	What is a priority encoder? Construct a 4-input priority encoder using combinational gates.	10	BOE310.2	K3

Course Outcome Wise Marks Distribution



Bloom's Level Distribution



Checked By
(Head of the Department)